The Brauer–Manin obstruction for curves

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Study group on the Brauer–Manin obstruction.

These notes have been written with the help of Claude Opus 5.5. I first asked Claude to conduct a research on the Brauer–Manin obstruction for curves, and then based on the results of this research, I structured the notes and asked Claude to fill in the details. I have then revised the notes and added some comments and references. The notes are not intended to be a complete reference on the topic, but rather a summary of the main ideas and results. I hope they will be useful for the study group and for anyone interested in the Brauer–Manin obstruction for curves.

1 Notation and preliminaries

The aim of this section is to fix the notation used throughout these notes and to recall the objects needed to state the Brauer–Manin obstruction and its relation with the Tate–Shafarevich group. Good general references are [Poo17], [Sko01] and [CTS21].

1.1 Global fields and adeles

Throughout, k denotes a global field, that is, either a number field or a global function field: the function field $k = \mathbb{F}_q(Y)$ of a nice curve Y over a finite field $\mathbb{F}_q$, where we always take $\mathbb{F}_q$ to be the full constant field of k (the algebraic closure of $\mathbb{F}_p$ in k). In the function field case, the places of k correspond to the closed points $y \in Y$, the completion at y is $k_y \simeq \mathbb{F}_{q^{\deg y}}((t))$, and there are no archimedean places. We fix a separable closure $\overline{k}$ of k and write $G_k = \mathrm{Gal}(\overline{k}/k)$ for the absolute Galois group.

- Ω_k denotes the set of places of k, and $\Omega_\infty \subseteq \Omega_k$ the (finite, possibly empty) subset of archimedean places.
- For $v \in \Omega_k$, k_v is the completion of k at v. If v is non-archimedean, $\mathcal{O}_v \subseteq k_v$ is its valuation ring.
- The ring of adeles of k is the restricted product with its usual locally compact topology:

$$\mathbb{A}_k = \prod_{v \in \Omega_k}{}' (k_v, \mathcal{O}_v) = \Big\{ (x_v)_v \in \prod_v k_v \ : \ x_v \in \mathcal{O}_v \text{ for all but finitely many } v \Big\}.$$

  The field k embeds diagonally in $\mathbb{A}_k$ as a discrete subgroup.

1.2 Varieties and adelic points

A nice variety over k is a smooth, projective, geometrically integral k-scheme. Unless stated otherwise, X will denote a nice variety and C a nice curve of genus g. For a field extension K/k we write $X_K = X \times_k K$, and $\overline{X} = X_{\overline{k}}$.

Since X is projective, its set of adelic points is simply

$$X(\mathbb{A}_k) = \prod_{v \in \Omega_k} X(k_v),$$

endowed with the product of the v-adic topologies. The diagonal map gives an inclusion $X(k)$ in $X(\mathbb{A}_k)$, so

$$X(k) \neq \emptyset \implies X(\mathbb{A}_k) \neq \emptyset.$$

Checking whether $X(\mathbb{A}_k) \neq \emptyset$ is a finite computation (Hensel's lemma and the Lang–Weil estimates deal with all but finitely many places), whereas deciding whether $X(k) \neq \emptyset$ is in general a very hard problem.

Definition 1.1. *Let $\mathcal{X}$ be a class of nice varieties over k.*

- *We say that $\mathcal{X}$ satisfies the Hasse principle if for every $X \in \mathcal{X}$, $X(\mathbb{A}_k) \neq \emptyset$ implies $X(k) \neq \emptyset$.*
- *We say that X satisfies weak approximation if $X(k)$ is dense in $X(\mathbb{A}_k)$.*

Example 1.2. *Conics (and more generally quadrics) satisfy the Hasse principle by the Hasse–Minkowski theorem. On the other hand, Selmer's cubic curve $3x^3 + 4y^3 + 5z^3 = 0$ in $\mathbb{P}^2_{\mathbb{Q}}$ has points over every $\mathbb{Q}_p$ and over $\mathbb{R}$, but no $\mathbb{Q}$-points.*

2 Brauer groups and the Brauer–Manin obstruction

Definition 2.1. *The Brauer group of a field K is*

$$\operatorname{Br} K = \mathrm{H}^2(G_K, \overline{K}^\times).$$

Equivalently, $\operatorname{Br} K$ is the group of Morita equivalence classes of central simple K-algebras, with the group law induced by $\otimes_K$. It is a torsion group.

Example 2.2. *For $a, b \in K^\times$ (and $\operatorname{char} K \neq 2$) the quaternion algebra (a, b) is the K-algebra generated by i, j with $i^2 = a$, $j^2 = b$, $ij = -ji$. Its class lies in $(\operatorname{Br} K)[2]$, and it is trivial if and only if the conic $ax^2 + by^2 = z^2$ has a K-point.*

The Brauer groups of local fields are computed by local class field theory: there are injective homomorphisms, the local invariants,

$$\mathrm{inv}_v \colon \operatorname{Br} k_v \hookrightarrow \mathbb{Q}/\mathbb{Z}, \qquad \text{with image} \quad \begin{cases} \mathbb{Q}/\mathbb{Z} & \text{if } v \text{ is non-archimedean,} \\ \frac{1}{2}\mathbb{Z}/\mathbb{Z} & \text{if } v \text{ is real,} \\ 0 & \text{if } v \text{ is complex.} \end{cases}$$

For instance, $\mathrm{inv}_v(a,b) = 0$ or $\frac{1}{2}$ according to whether the Hilbert symbol $(a,b)_v$ is 1 or -1.

Theorem 2.3 (Albert–Brauer–Hasse–Noether)**.** *There is an exact sequence*

$$0 \longrightarrow \operatorname{Br} k \longrightarrow \bigoplus_{v\in\Omega_k} \operatorname{Br} k_v \xrightarrow{\sum_v \mathrm{inv}_v} \mathbb{Q}/\mathbb{Z} \longrightarrow 0.$$

In particular, for every $\alpha \in \operatorname{Br} k$, $\mathrm{inv}_v(\alpha_v) = 0$ *for all but finitely many* v, *and* $\sum_v \mathrm{inv}_v(\alpha_v) = 0$.

The second statement (the *reciprocity law*) is what makes the Brauer–Manin obstruction work; it is a reformulation of Hilbert's reciprocity law $\prod_v (a,b)_v = 1$ in the quaternion case.

2.1 The Brauer group of a variety

Definition 2.4. *The (cohomological)* Brauer group *of a scheme* X *is* $\operatorname{Br} X = \mathrm{H}^2_{\mathrm{et}}(X, \mathbb{G}_{\mathrm{m}})$.

For X regular and integral (e.g. X nice), $\operatorname{Br} X$ is a torsion group which injects into $\operatorname{Br} k(X)$, where $k(X)$ is the function field of X. The construction $X \mapsto \operatorname{Br} X$ is a contravariant functor. In particular, for any field extension K/k and any $P \in X(K)$, i.e. $P\colon \operatorname{Spec} K \to X$, we get the evaluation map

$$\begin{aligned} \operatorname{Br} X &\longrightarrow \operatorname{Br} K, \\ \alpha &\longmapsto \alpha(P) := P^*\alpha. \end{aligned}$$

We use the standard filtration

$$\operatorname{Br}_0 X \subseteq \operatorname{Br}_1 X \subseteq \operatorname{Br} X, \qquad \operatorname{Br}_0 X = \operatorname{im}\big(\operatorname{Br} k \to \operatorname{Br} X\big), \qquad \operatorname{Br}_1 X = \ker\big(\operatorname{Br} X \to \operatorname{Br} \overline{X}\big).$$

Elements of $\operatorname{Br}_1 X$ are called algebraic, and those of $\operatorname{Br} X \backslash \operatorname{Br}_1 X$ transcendental. Since $\mathrm{H}^3(k, \overline{k}^\times) = 0$ for a global field k, the Hochschild–Serre spectral sequence gives an exact sequence

$$0 \longrightarrow \operatorname{Pic} X \longrightarrow (\operatorname{Pic} \overline{X})^{G_k} \longrightarrow \operatorname{Br} k \longrightarrow \operatorname{Br}_1 X \longrightarrow \mathrm{H}^1(k, \operatorname{Pic} \overline{X}) \longrightarrow 0, \tag{2.5}$$

and hence an isomorphism $\operatorname{Br}_1 X/\operatorname{Br}_0 X \simeq \mathrm{H}^1(k, \operatorname{Pic} \overline{X})$. If moreover $X(\mathbb{A}_k) \neq \emptyset$, then $\operatorname{Br} k \to \operatorname{Br} X$ is injective (by Theorem 2.3, as $\operatorname{Br} k_v \to \operatorname{Br} X_{k_v}$ has a retraction given by any $P_v \in X(k_v)$), and so $\operatorname{Pic} X = (\operatorname{Pic} \overline{X})^{G_k}$.

Theorem 2.6 (Tsen [Tse33])**.** *Let* C *be a nice curve over* k. *Then* $\operatorname{Br} \overline{C} = 0$. *Consequently* $\operatorname{Br} C = \operatorname{Br}_1 C$, *that is,* every Brauer class on a curve is algebraic, *and*

$$\operatorname{Br} C/\operatorname{Br}_0 C \simeq \mathrm{H}^1(k, \operatorname{Pic} \overline{C}).$$

Proof. Tsen's theorem states that the function field $\overline{k}(C)$ of a curve over an algebraically closed field is C_1, which implies $\operatorname{Br} \overline{k}(C) = 0$. Since $\overline{C}$ is regular and integral, $\operatorname{Br} \overline{C} \hookrightarrow \operatorname{Br} \overline{k}(C) = 0$ (see [CTS21, Ch. 5]). The last statement then follows from (2.5). □

2.2 The Brauer–Manin obstruction

Lemma 2.7. *Let $\alpha \in \operatorname{Br} X$. Then:*

1. *for each v, the map $X(k_v) \to \operatorname{Br} k_v$, $P_v \mapsto \alpha(P_v)$, is locally constant;*
2. *there is a finite set $S \subseteq \Omega_k$ such that $\alpha(P_v) = 0$ for all $v \notin S$ and all $P_v \in X(k_v)$.*

The second part follows by spreading α out to a smooth proper model $\mathcal{X}$ of X over the S-integers, using that $\operatorname{Br} \mathcal{O}_v = 0$ and that $X(k_v) = \mathcal{X}(\mathcal{O}_v)$ for $v \notin S$. Lemma 2.7 shows that the following pairing is well defined and continuous in the first variable.

Definition 2.8. *The Brauer–Manin pairing is*

$$X(\mathbb{A}_k) \times \operatorname{Br} X \longrightarrow \mathbb{Q}/\mathbb{Z}, \qquad \big\langle (P_v)_v, \alpha \big\rangle = \sum_{v \in \Omega_k} \operatorname{inv}_v \alpha(P_v).$$

For a subset $B \subseteq \operatorname{Br} X$ we define

$$X(\mathbb{A}_k)^B = \big\{ (P_v)_v \in X(\mathbb{A}_k) \ : \ \langle (P_v)_v, \alpha \rangle = 0 \text{ for all } \alpha \in B \big\},$$

and the Brauer set is $X(\mathbb{A}_k)^{\mathrm{Br}} := X(\mathbb{A}_k)^{\operatorname{Br} X}$.

By Theorem 2.3, elements of $\operatorname{Br}_0 X$ pair to zero with every adelic point, so the pairing descends to $X(\mathbb{A}_k) \times \operatorname{Br} X/\operatorname{Br}_0 X \to \mathbb{Q}/\mathbb{Z}$. Applying Theorem 2.3 to $\alpha(P) \in \operatorname{Br} k$ for $P \in X(k)$, and using that $X(\mathbb{A}_k)^{\mathrm{Br}}$ is closed, we obtain

$$X(k) \subseteq \overline{X(k)} \subseteq X(\mathbb{A}_k)^{\mathrm{Br}} \subseteq X(\mathbb{A}_k),$$

where $\overline{X(k)}$ denotes the topological closure.

Definition 2.9. *We say that there is a Brauer–Manin obstruction to the Hasse principle for X if $X(\mathbb{A}_k) \neq \emptyset$ but $X(\mathbb{A}_k)^{\mathrm{Br}} = \emptyset$, and a Brauer–Manin obstruction to weak approximation if $X(\mathbb{A}_k)^{\mathrm{Br}} \neq X(\mathbb{A}_k)$. We say that the Brauer–Manin obstruction is the only obstruction (to the Hasse principle) for X if $X(\mathbb{A}_k)^{\mathrm{Br}} \neq \emptyset$ implies $X(k) \neq \emptyset$.*

3 The conjecture for curves

The shape of the set $C(k)$ we are trying to cut out depends heavily on the genus g of C.

We have the following classical results, which are the starting point of the study of rational points on curves.

Theorem 3.1 (Minkowski [Min90]; Hasse [Has23, Has24]). *Let C be a nice curve of genus 0 over a global field k. Then C is isomorphic to a conic in $\mathbb{P}^2_k$, and:*

1. *C satisfies the Hasse principle: if $C(\mathbb{A}_k) \neq \emptyset$, then $C(k) \neq \emptyset$;*
2. *if $C(k) \neq \emptyset$, then $C \simeq \mathbb{P}^1_k$, so $C(k)$ is infinite, and C satisfies weak approximation, i.e. $C(k)$ is dense in $C(\mathbb{A}_k)$.*

Theorem 3.2 (Mordell [Mor22]; Weil [Wei29]; Lang–Néron [LN59]). *Let C be a nice curve of genus 1 over a global field k. If $C(k) \neq \emptyset$, then any choice of $O \in C(k)$ makes (C, O) an elliptic curve E, and $C(k) = E(k)$ is a finitely generated abelian group,*

$$E(k) \simeq \mathbb{Z}^r \oplus E(k)_{\mathrm{tors}}, \qquad E(k)_{\mathrm{tors}} \text{ finite.}$$

Minkowski proved the local-global principle for quadratic forms over $\mathbb{Q}$ in 1890. Hasse reproved it in 1923 and extended it to number fields in 1924. Over any global field, the Hasse principle for conics also follows from the injectivity of $\operatorname{Br} k \to \bigoplus_v \operatorname{Br} k_v$ in Theorem 2.3, since a conic has a k-point if and only if the class of the corresponding quaternion algebra is trivial. Mordell proved Theorem 3.2 for $k = \mathbb{Q}$ in 1922, and Weil extended it to number fields (and to Jacobians of curves of any genus) in 1929. Over a global function field, the Lang–Néron theorem (1959) says that $E(k)$ is finitely generated modulo the image of the $k/\mathbb{F}_q$-trace of E; since the constant field $\mathbb{F}_q$ is finite, $E(k)$ is finitely generated.

So in genus 1 the set $C(k)$ may be empty, finite (if $r = 0$) or infinite (if $r > 0$). Unlike in genus 0, the Hasse principle can fail, as Selmer's cubic shows, and deciding whether $C(k) \neq \emptyset$ is already a difficult problem, closely tied to the Tate–Shafarevich group of E (see Section 4.4). For curves of genus at least 2, $C(k)$ is finite. Recall that a curve C over a global function field k is isotrivial if $C_{\overline{k}}$ is isomorphic to the base change of a curve defined over a finite field.

Theorem 3.3 (Faltings [Fal83]; Manin [Man63], Grauert [Gra65], Samuel [Sam66]). *Let C be a nice curve of genus $g \geq 2$ over a global field k.*

1. *If k is a number field, then $C(k)$ is finite.*
2. *If k is a global function field and C is not isotrivial, then $C(k)$ is finite.*

The first part is the Mordell conjecture, proved by Faltings in 1983. The function field analogue was proved earlier: by Manin and Grauert in characteristic 0, and by Samuel in characteristic p for non-isotrivial curves. The isotriviality assumption cannot be dropped: if C_0 is a nice curve of genus ≥ 2 over $\mathbb{F}_q$ and $k = \mathbb{F}_q(C_0)$, the constant curve $C = C_0 \times_{\mathbb{F}_q} k$ has the infinitely many k-points given by the generic point of C_0 composed with the powers of the q-Frobenius $C_0 \to C_0$.

For curves of genus $g \geq 2$, one would like the Brauer set to cut out exactly the finite set $C(k)$. Over a number field, this cannot be true as stated.

Suppose $g \geq 2$, k has a real place v_0 and $C(k) \neq \emptyset$. By Lemma 2.7, each $\alpha \in \operatorname{Br} C$ is constant on the connected components of $C(k_{v_0})$. So if $(P_v)_v \in C(\mathbb{A}_k)^{\mathrm{Br}}$, then replacing P_{v_0} by any other point on the same connected component of $C(k_{v_0})$ produces another element of $C(\mathbb{A}_k)^{\mathrm{Br}}$. Hence $C(\mathbb{A}_k)^{\mathrm{Br}}$ contains whole real components, which are uncountable, and it can *never* be equal to the finite set $C(k)$ (see Figure 1).

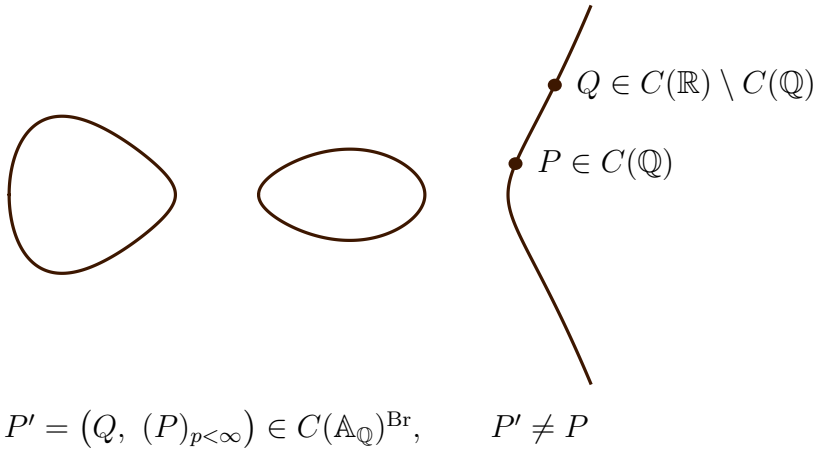


Figure 1: *The real points of the genus-2 curve C over $\mathbb{Q}$, which has three connected components. If $P \in C(\mathbb{Q})$ and Q is any real point on the same component as P, the Brauer–Manin pairing cannot distinguish P' from P, so $P' \in C(\mathbb{A}_{\mathbb{Q}})^{\mathrm{Br}}$ even though it is not rational. In $C(\mathbb{A}_{\mathbb{Q}})_\bullet$, the points P' and P become equal.*

Modified adelic points. Since the Brauer group cannot distinguish points on the same connected component at an archimedean place, these are exactly the points one should identify. Following Poonen and Stoll [Sto07], we define

$$X(\mathbb{A}_k)_\bullet = \prod_{v \notin \Omega_\infty} X(k_v) \times \prod_{v \in \Omega_\infty} \pi_0\big(X(k_v)\big),$$

where π_0 denotes the set of connected components. For a complex place v, $X(k_v)$ is connected, so the corresponding factor is a single point, and if k is a function field, $X(\mathbb{A}_k)_\bullet = X(\mathbb{A}_k)$. There is a natural continuous surjection $X(\mathbb{A}_k) \twoheadrightarrow X(\mathbb{A}_k)_\bullet$, and we still have a diagonal map $X(k) \to X(\mathbb{A}_k)_\bullet$. Since the Brauer–Manin pairing factors through $X(\mathbb{A}_k)_\bullet$, we define $X(\mathbb{A}_k)_\bullet^B$ and $X(\mathbb{A}_k)_\bullet^{\mathrm{Br}}$ analogously, and as before $X(k) \subseteq \overline{X(k)} \subseteq X(\mathbb{A}_k)_\bullet^{\mathrm{Br}}$. Moreover, the usual Brauer set is exactly the preimage of the modified one:

$$X(\mathbb{A}_k)^{\mathrm{Br}} = \Big\{(P_v)_v \in X(\mathbb{A}_k) \ :\ \text{its image in } X(\mathbb{A}_k)_\bullet \text{ lies in } X(\mathbb{A}_k)_\bullet^{\mathrm{Br}}\Big\}.$$

So passing to $X(\mathbb{A}_k)_\bullet$ loses no information about the Brauer–Manin obstruction, but it makes it possible to state the strongest conjecture one can hope for.

The question underlying these notes is:

Conjecture 3.4 (Scharaschkin, Skorobogatov)**.** *Let C be a nice curve over a number field k. Then $C(\mathbb{A}_k)_\bullet^{\mathrm{Br}} = \overline{C(k)}$, where the closure is taken in $C(\mathbb{A}_k)_\bullet$. In particular, if $g \geq 2$ (so $C(k)$ is finite by Theorem 3.3), then $C(\mathbb{A}_k)_\bullet^{\mathrm{Br}} = C(k)$.*

In concrete terms, Conjecture 3.4 for $g \geq 2$ says that an element of $C(\mathbb{A}_k)^{\mathrm{Br}}$ coincides with some $P \in C(k)$ at every non-archimedean place, and lies on the same connected component as P at every real place.

Two cases where the conjecture is known. We discuss further cases in Section 4.4, once the Jacobian and the Tate–Shafarevich group have been introduced. Two cases need nothing more than what we have so far.

- *Genus* 0. Here C is a conic. If $C(\mathbb{A}_k) = \emptyset$ there is nothing to prove. Otherwise $C \simeq \mathbb{P}^1$ by the Hasse–Minkowski theorem, so $\operatorname{Br} C = \operatorname{Br}_0 C$, and therefore $C(\mathbb{A}_k)^{\mathrm{Br}} = C(\mathbb{A}_k)$. Weak approximation for $\mathbb{P}^1$ gives $C(\mathbb{A}_k) = \overline{C(k)}$. So Conjecture 3.4 holds unconditionally in genus 0.

- *Global function fields.* The analogue of Conjecture 3.4 makes sense over a global function field k. Since k has no archimedean places, no modification of the adelic points is needed. For curves of genus $g \geq 2$, the analogue is now a theorem, except in the isotrivial case.

  Theorem 3.5 (Poonen–Voloch [PV10]; Creutz–Voloch [CV26])**.** *Let C be a nice curve of genus $g \geq 2$ over a global function field k. If C is not isotrivial, then*

  $$C(\mathbb{A}_k)^{\mathrm{Br}} = C(k).$$

  Poonen and Voloch proved this for curves (and, more generally, for coset-free subvarieties) contained in an abelian variety A such that $A_{\overline{k}}$ has no non-zero isotrivial quotient and $A(k^{\mathrm{sep}})[p]$ is finite, where $p = \operatorname{char} k$. Creutz and Voloch removed all hypotheses on the ambient abelian variety. In both cases, the key input is an adelic intersection statement,

  $$C(\mathbb{A}_k) \cap \overline{A(k)} = C(k) \qquad \text{inside } A(\mathbb{A}_k),$$

  together with a variant using Selmer groups, which links it to the Brauer set. The analogue of this intersection statement over number fields is wide open.

  The isotrivial case is genuinely different. As we saw after Theorem 3.3, a constant curve can have infinitely many k-points, so one can only hope for $C(\mathbb{A}_k)^{\mathrm{Br}} = \overline{C(k)}$ (see [PV10, Remark 1.3]). For constant curves, the following is known.

  Theorem 3.6 (Creutz–Voloch [CV22])**.** *Let C_0 and D be nice curves over a finite field $\mathbb{F}$, let $k = \mathbb{F}(D)$, and let $C = C_0 \times_{\mathbb{F}} k$. If the genus of D is smaller than the genus of C_0, then the Brauer–Manin obstruction is the only obstruction to the Hasse principle and to weak approximation for C, i.e. $C(\mathbb{A}_k)^{\mathrm{Br}} = \overline{C(k)}$.*

  The remaining isotrivial cases are open.

4 Jacobians, torsors and the Tate–Shafarevich group

4.1 Curves and their Jacobians

From now on, k denotes a number field. Much of what follows holds over any global field, but over a global function field of characteristic p one has to replace Galois cohomology by flat cohomology when dealing with p-primary parts (of Selmer groups, of Ш, and of the Cassels–Tate pairing), and we will not discuss this.

The Picard scheme. We want a variety over k whose points are line bundles on C, even when C has no rational points. The naive functor $T \mapsto \operatorname{Pic}(C \times_k T)/\operatorname{Pic}(T)$ is not representable in general. Its sheafification for the étale topology, however, is represented by a k-scheme $\mathbf{Pic}_{C/k}$, locally of finite type, called the Picard scheme of C (Grothendieck; see [Kle05] or [Poo17]). It decomposes according to the degree as

$$\mathbf{Pic}_{C/k} = \bigsqcup_{d\in\mathbb{Z}} \mathbf{Pic}^d_{C/k}\,.$$

Because of the sheafification, a k-point of $\mathbf{Pic}_{C/k}$ only needs to be a line bundle on $\overline{C}$ whose class is fixed by Galois:

$$\mathbf{Pic}_{C/k}(\overline{k}) = \operatorname{Pic}\overline{C}, \qquad \mathbf{Pic}_{C/k}(k) = (\operatorname{Pic}\overline{C})^{G_k}.$$

The latter can be strictly larger than $\operatorname{Pic} C$, the group of classes of k-rational divisors. By (2.5), the difference is measured by $\operatorname{Br} k$, and it vanishes when $C(\mathbb{A}_k) \neq \emptyset$.

Example 4.1. *Let C be a conic over k with $C(k) = \emptyset$. Over $\overline{k}$ we have $\overline{C} \simeq \mathbb{P}^1$, so all points of $C(\overline{k})$ are linearly equivalent. Hence the class of any $Q \in C(\overline{k})$ is Galois invariant, and $\mathbf{Pic}^1_{C/k}(k) \neq \emptyset$. However, every k-rational divisor on C has even degree, so this class is not represented by any k-rational divisor. Its image in $\operatorname{Br} k$ under (2.5) is the class of the quaternion algebra associated with C.*

Now let C be a nice curve of genus $g \geq 1$ over k and let $J = \mathbf{Pic}^0_{C/k}$ be its Jacobian, an abelian variety of dimension g with $J(\overline{k}) = \operatorname{Pic}^0\overline{C}$ and $J(k) = (\operatorname{Pic}^0\overline{C})^{G_k}$. The degree map gives an exact sequence of G_k-modules

$$0 \longrightarrow J(\overline{k}) \longrightarrow \operatorname{Pic}\overline{C} \xrightarrow{\deg} \mathbb{Z} \longrightarrow 0. \tag{4.2}$$

- The degree-one component $\mathbf{Pic}^1_{C/k}$ is a torsor under J (see below). It has a k-point if and only if C has a k-rational divisor class of degree 1, i.e. an element of $(\operatorname{Pic}\overline{C})^{G_k}$ of degree 1. This holds if $C(k) \neq \emptyset$, but can also hold when $C(k) = \emptyset$. If $C(\mathbb{A}_k) \neq \emptyset$, such a class is represented by a k-rational divisor, as noted above.

- Given a rational divisor class D of degree 1, we have the Abel–Jacobi map which is a closed immersion if $g \geq 1$.
$$\begin{aligned} \iota \colon C &\longrightarrow J, \\ P &\longmapsto [P - D]. \end{aligned}$$
It induces maps $C(k) \to J(k)$, $C(\mathbb{A}_k) \to J(\mathbb{A}_k)$ and $C(\mathbb{A}_k)_\bullet \to J(\mathbb{A}_k)_\bullet$ compatible with the diagonal embeddings.

Taking Galois cohomology in (4.2) relates the Brauer group of C to torsors under J.

Proposition 4.3. *Let C be a nice curve of genus $g \geq 1$ over k, with Jacobian J. Then*
$$\operatorname{Br} C / \operatorname{Br}_0 C \simeq \mathrm{H}^1(k, J) / \big\langle [\mathbf{Pic}^1_{C/k}] \big\rangle. \tag{4.4}$$
In particular, $\operatorname{Br} C / \operatorname{Br}_0 C \simeq \mathrm{H}^1(k, J)$ *whenever C has a k-rational divisor class of degree* 1.

Proof. The sequence (4.2) is a sequence of G_k-modules, where G_k acts on $\operatorname{Pic} \overline{C}$ through its action on $C(\overline{k})$ and trivially on $\mathbb{Z}$: the degree map is Galois equivariant because $\sigma \in G_k$ sends points to points. It is surjective, since $\deg[Q] = 1$ for any $Q \in C(\overline{k})$. Its long exact sequence in Galois cohomology reads
$$0 \longrightarrow J(k) \longrightarrow (\operatorname{Pic} \overline{C})^{G_k} \xrightarrow{\deg} \mathbb{Z} \xrightarrow{\delta} \mathrm{H}^1(k, J) \longrightarrow \mathrm{H}^1(k, \operatorname{Pic} \overline{C}) \longrightarrow \mathrm{H}^1(k, \mathbb{Z}),$$
where we used $\mathbb{Z}^{G_k} = \mathbb{Z}$.

Step 1: $\mathrm{H}^1(k, \mathbb{Z}) = 0$. Since G_k acts trivially on $\mathbb{Z}$, 1-cocycles are continuous homomorphisms and coboundaries are zero, so $\mathrm{H}^1(k, \mathbb{Z}) = \operatorname{Hom}_{\mathrm{cont}}(G_k, \mathbb{Z})$. The image of such a homomorphism is a compact, hence finite, subgroup of the discrete group $\mathbb{Z}$, so it is zero. Therefore
$$\mathrm{H}^1(k, \operatorname{Pic} \overline{C}) \simeq \mathrm{H}^1(k, J) / \operatorname{im} \delta.$$

Step 2: $\delta(1) = [\mathbf{Pic}^1_{C/k}]$. To compute $\delta(1)$, lift $1 \in \mathbb{Z}$ to $[Q] \in \operatorname{Pic} \overline{C}$ for some $Q \in C(\overline{k})$. Then $\delta(1)$ is the class of the cocycle
$$\sigma \longmapsto \sigma[Q] - [Q] = [\sigma Q - Q] \in J(\overline{k}).$$
On the other hand, if T is a torsor under J and $x \in T(\overline{k})$, the class of T in $\mathrm{H}^1(k, J)$ is represented by $\sigma \mapsto \sigma(x) - x$. Applying this to $T = \mathbf{Pic}^1_{C/k}$ and $x = [Q] \in \mathbf{Pic}^1_{C/k}(\overline{k})$ shows that $\delta(1) = [\mathbf{Pic}^1_{C/k}]$. Since $\mathbb{Z}$ is generated by 1, $\operatorname{im} \delta = \big\langle [\mathbf{Pic}^1_{C/k}] \big\rangle$.

Step 3. By Theorem 2.6, $\operatorname{Br} C / \operatorname{Br}_0 C \simeq \mathrm{H}^1(k, \operatorname{Pic} \overline{C})$, and (4.4) follows from Steps 1 and 2. Finally, by exactness at $\mathbb{Z}$, $\delta(1) = 0$ if and only if 1 lies in the image of $\deg \colon (\operatorname{Pic} \overline{C})^{G_k} \to \mathbb{Z}$, that is, if and only if C has a k-rational divisor class of degree 1. In that case the quotient in (4.4) is trivial. □

Thus, on a curve, Brauer classes are essentially the same thing as torsors under the Jacobian. This is the reason why the Brauer–Manin obstruction on curves is governed by the arithmetic of J.

4.2 Torsors and the Tate–Shafarevich group

Let A be an abelian variety over k and $A^\vee$ its dual abelian variety. For a Jacobian, $J^\vee \simeq J$ via the canonical principal polarisation. By the Mordell–Weil theorem (see the discussion after Theorem 3.2), $A(k)$ is finitely generated. We write $\overline{A(k)}$ for the closure of $A(k)$ in $A(\mathbb{A}_k)_\bullet$. It is in general strictly larger than $A(k)$, and it will play a central role in Section 5.

Definition 4.5. *A torsor (or principal homogeneous space) under A is a variety T over k with an action $A \times T \to T$ such that $T_{\overline{k}}$, with its $A_{\overline{k}}$-action, is isomorphic to $A_{\overline{k}}$ acting on itself by translations.*

Isomorphism classes of torsors are classified by the Weil–Châtelet group

$$\mathrm{H}^1(k, A) := \mathrm{H}^1\big(G_k, A(\overline{k})\big),$$

and the class of T is trivial if and only if $T(k) \neq \emptyset$ (in which case $T \simeq A$). For example, every nice genus-one curve C is a torsor under its Jacobian elliptic curve $E = \mathbf{Pic}^0_{C/k}$, via $C \simeq \mathbf{Pic}^1_{C/k}$.

Definition 4.6. *The Tate–Shafarevich group of A is*

$$Ш(k, A) = \ker\Big(\mathrm{H}^1(k, A) \longrightarrow \prod_{v \in \Omega_k} \mathrm{H}^1(k_v, A)\Big).$$

Concretely, $Ш(k, A)$ is the group of torsors T under A with $T(\mathbb{A}_k) \neq \emptyset$, and such a torsor represents a non-zero element if and only if $T(k) = \emptyset$. In other words, *non-zero elements of $Ш(k, A)$ are exactly the torsors under A that violate the Hasse principle.*

Example 4.7. *Selmer's curve $3x^3+4y^3+5z^3 = 0$ is a torsor under its Jacobian $E\colon x^3+y^3+60z^3 = 0$, and represents a non-zero element of $Ш(\mathbb{Q}, E)[3]$.*

Selmer groups. Let $n \geq 1$. The Kummer sequence $0 \to A[n] \to A \xrightarrow{n} A \to 0$ gives, over k and over every k_v, a short exact sequence

$$0 \longrightarrow A(k)/nA(k) \xrightarrow{\delta} \mathrm{H}^1(k, A[n]) \longrightarrow \mathrm{H}^1(k, A)[n] \longrightarrow 0.$$

These sequences are compatible with restriction from k to each k_v, so we get a commutative diagram with exact rows

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A(k)/nA(k) & \xrightarrow{\delta} & \mathrm{H}^1(k, A[n]) & \longrightarrow & \mathrm{H}^1(k, A)[n] & \longrightarrow & 0 \\ & & \downarrow & & \downarrow{\scriptstyle \mathrm{res}} & & \downarrow{\scriptstyle \mathrm{res}} & & \\ 0 & \longrightarrow & \prod_v A(k_v)/nA(k_v) & \xrightarrow{(\delta_v)_v} & \prod_v \mathrm{H}^1(k_v, A[n]) & \longrightarrow & \prod_v \mathrm{H}^1(k_v, A)[n] & \longrightarrow & 0. \end{array} \tag{4.8}$$

Definition 4.9. *The n-Selmer group of A is*

$$\mathrm{Sel}^{(n)}(k, A) = \big\{\xi \in \mathrm{H}^1(k, A[n]) \ : \ \xi_v \in \delta_v\big(A(k_v)/nA(k_v)\big) \text{ for all } v \in \Omega_k\big\}.$$

By exactness of the bottom row, ξ_v lies in the image of δ_v if and only if it maps to 0 in $\mathrm{H}^1(k_v, A)$. So $\mathrm{Sel}^{(n)}(k, A)$ is the kernel of the composite map $\mathrm{H}^1(k, A[n]) \to \prod_v \mathrm{H}^1(k_v, A)[n]$ in (4.8). It is a finite, effectively computable group, sitting in the fundamental exact sequence

$$0 \longrightarrow A(k)/nA(k) \longrightarrow \mathrm{Sel}^{(n)}(k, A) \longrightarrow Ш(k, A)[n] \longrightarrow 0. \tag{4.10}$$

This follows from a diagram chase in (4.8):

- By commutativity of the right-hand square, an element $\xi \in \mathrm{Sel}^{(n)}(k, A)$ maps to a class in $\mathrm{H}^1(k, A)[n]$ that is locally trivial everywhere, i.e. to an element of $Ш(k, A)[n]$.
- This map is surjective: given $t \in Ш(k, A)[n]$, lift it to some $\xi \in \mathrm{H}^1(k, A[n])$ using the top row. The image of ξ in $\prod_v \mathrm{H}^1(k_v, A)[n]$ is the image of t, which is 0. Hence $\xi \in \mathrm{Sel}^{(n)}(k, A)$.
- By commutativity of the left-hand square, $\delta(A(k)/nA(k)) \subseteq \mathrm{Sel}^{(n)}(k, A)$. By exactness of the top row, this is exactly the kernel of $\mathrm{Sel}^{(n)}(k, A) \to Ш(k, A)[n]$, and δ is injective.

Since $\mathrm{H}^1(k, A)$ is torsion (every class is killed by the degree of a finite extension over which it becomes trivial), so is $Ш(k, A)$; and by (4.10), $Ш(k, A)[n]$ is finite for every n. Taking inverse limits over n (ordered by divisibility) we get the profinite Selmer group $\widehat{\mathrm{Sel}}(k, A) = \varprojlim_n \mathrm{Sel}^{(n)}(k, A)$ and an exact sequence

$$0 \longrightarrow \widehat{A(k)} \longrightarrow \widehat{\mathrm{Sel}}(k, A) \longrightarrow T Ш(k, A) \longrightarrow 0,$$

Here

$$\widehat{A(k)} = \varprojlim_n A(k)/nA(k) \qquad \text{and} \qquad T Ш(k, A) = \varprojlim_n Ш(k, A)[n]$$

are the profinite completion of $A(k)$ and the Tate module of $Ш$, respectively.

The divisible part. An abelian group G is divisible if every element can be divided by every positive integer, i.e. for all $x \in G$ and $n \geq 1$ there is $y \in G$ with $ny = x$. Equivalently, multiplication by n is surjective on G for every $n \geq 1$. Typical examples are $\mathbb{Q}$, $\mathbb{Q}/\mathbb{Z}$ and $\mathbb{Q}_p/\mathbb{Z}_p$, while a non-zero finite group is never divisible. The sum of two divisible subgroups is again divisible, so every abelian group has a unique maximal divisible subgroup. Let $Ш(k, A)_{\mathrm{div}}$ be the maximal divisible subgroup of $Ш(k, A)$. Since all the $Ш(k, A)[n]$ are finite, one has

$$T Ш(k, A) = 0 \iff Ш(k, A)_{\mathrm{div}} = 0.$$

Conjecture 4.11 (Tate–Shafarevich)**.** *$Ш(k, A)$ is finite. In particular, $Ш(k, A)_{\mathrm{div}} = 0$.*

This is known only in special cases, e.g. for elliptic curves over $\mathbb{Q}$ of analytic rank at most 1 (Kolyvagin, Gross–Zagier). Finiteness of $Ш$, or rather the weaker condition $Ш_{\mathrm{div}} = 0$, is the hypothesis that will appear in the main theorems of these notes.

4.3 The Cassels–Tate pairing and Manin's theorem

The question. A non-zero element of $Ш(k, A)$ is a torsor T under A with $T(\mathbb{A}_k) \neq \emptyset$ but $T(k) = \emptyset$, that is, a counterexample to the Hasse principle. It is natural to ask whether this failure is explained by the Brauer–Manin obstruction: is there some $\alpha \in \operatorname{Br} T$ such that $\langle (P_v)_v, \alpha \rangle \neq 0$ for *every* $(P_v)_v \in T(\mathbb{A}_k)$? Since $T(\mathbb{A}_k)$ is large, the simplest way for this to happen is for the pairing with α not to depend on the adelic point at all. This leads us to consider the following group.

Manin's group. Define

$$Б(X) = \ker\Big(\operatorname{Br}_1 X/\operatorname{Br}_0 X \longrightarrow \prod_{v\in\Omega_k} \operatorname{Br}_1 X_{k_v}/\operatorname{Br}_0 X_{k_v}\Big) \simeq Ш^1\big(k, \operatorname{Pic}\overline{X}\big),$$

where, for a G_k-module M, $Ш^1(k, M) = \ker\big(\mathrm{H}^1(k, M) \to \prod_v \mathrm{H}^1(k_v, M)\big)$, and the isomorphism comes from (2.5) over k and over every k_v. If $\alpha \in \operatorname{Br}_1 X$ represents an element of $Б(X)$, then for each v its restriction to X_{k_v} lies in $\operatorname{Br}_0 X_{k_v}$, i.e. it is the pull-back of some $c_v \in \operatorname{Br} k_v$. Evaluating such a class at any $P_v \in X(k_v)$ just returns c_v, because $\operatorname{Spec} k_v \xrightarrow{P_v} X_{k_v} \to \operatorname{Spec} k_v$ is the identity. Hence

$$\big\langle (P_v)_v, \alpha \big\rangle = \sum_{v\in\Omega_k} \operatorname{inv}_v c_v =: c(\alpha) \in \mathbb{Q}/\mathbb{Z}$$

is *constant* on $X(\mathbb{A}_k)$. It only depends on the class of α modulo $\operatorname{Br}_0 X$, since changing α by $b \in \operatorname{Br} k$ changes each c_v by b_v, and $\sum_v \operatorname{inv}_v b_v = 0$ by Theorem 2.3. If $c(\alpha) \neq 0$, then $X(\mathbb{A}_k)^{\mathrm{Br}} = \emptyset$. Thus $Б(X)$ can only obstruct the Hasse principle, never weak approximation.

Locally constant classes on torsors. Let T be a torsor under A. Over $\overline{k}$ we have $\overline{T} \simeq \overline{A}$, and since translations act trivially on Pic^0, there is an isomorphism of G_k-modules $\operatorname{Pic}^0 \overline{T} \simeq \operatorname{Pic}^0 \overline{A} = A^\vee(\overline{k})$. The inclusion $A^\vee(\overline{k}) \subseteq \operatorname{Pic}\overline{T}$ induces a map $\mathrm{H}^1(k, A^\vee) \to \mathrm{H}^1(k, \operatorname{Pic}\overline{T})$, and similarly over every k_v. It therefore sends locally trivial classes to locally trivial classes, giving

$$Ш(k, A^\vee) \longrightarrow Ш^1\big(k, \operatorname{Pic}\overline{T}\big) \simeq Б(T), \qquad \eta \longmapsto \beta_\eta.$$

So *every* element of $Ш(k, A^\vee)$ gives a candidate obstruction on *every* torsor T. Recording the constant values yields a pairing

$$Ш(k, A) \times Ш(k, A^\vee) \longrightarrow \mathbb{Q}/\mathbb{Z}, \qquad \big([T], \eta\big) \longmapsto c(\beta_\eta) = \sum_{v\in\Omega_k} \operatorname{inv}_v \beta_\eta(P_v), \quad (P_v)_v \in T(\mathbb{A}_k).$$

The question above becomes: given $[T] \neq 0$, is there an η that pairs non-trivially with it?

The Cassels–Tate pairing. The pairing just constructed is a classical object. Cassels (for elliptic curves) and Tate (in general) defined, by purely Galois-cohomological means, a bilinear pairing

$$\langle\ ,\ \rangle_{\mathrm{CT}} \colon Ш(k, A) \times Ш(k, A^\vee) \longrightarrow \mathbb{Q}/\mathbb{Z}$$

whose left and right kernels are exactly the maximal divisible subgroups. Manin [Man71] showed that, up to sign,

$$c(\beta_\eta) = \langle [T], \eta \rangle_{\mathrm{CT}}.$$

In other words, *the Cassels–Tate pairing is the Brauer–Manin pairing against the locally constant classes.* If $Ш(k, A)$ is finite, then so is $Ш(k, A^\vee)$, since A and $A^\vee$ are isogenous, and the pairing is perfect. This gives the following.

Theorem 4.12 (Manin [Man71])**.** *Let T be a torsor under an abelian variety A over a number field k such that $Ш(k, A)$ is finite. Then the Brauer–Manin obstruction is the only obstruction to the Hasse principle for T. In particular, this holds for genus-one curves C whose Jacobian has finite $Ш$.*

Proof. We sketch the proof for genus-one curves. Let C be a nice curve of genus 1 over k with Jacobian E, and suppose that $Ш(k, E)$ is finite, $C(\mathbb{A}_k) \neq \emptyset$ and $C(k) = \emptyset$. We want a *single* Brauer class β on C that pairs non-trivially with *every* adelic point. The natural candidates are the locally constant classes β_η constructed above, whose pairing does not depend on the point at all, so it suffices to show that one of these constants is non-zero.

Step 1: C gives a non-zero element of $Ш(k, E)$. The curve $C \simeq \mathbf{Pic}^1_{C/k}$ is a torsor under E, and a torsor is trivial if and only if it has a rational point. Since $C(k_v) \neq \emptyset$ for every v, each $[C]_v \in \mathrm{H}^1(k_v, E)$ is zero, so $[C] \in Ш(k, E)$. Since $C(k) = \emptyset$, $[C] \neq 0$.

Step 2: choose η with $\langle [C], \eta \rangle_{\mathrm{CT}} \neq 0$. A finite group has no non-zero divisible subgroup, so the Cassels–Tate pairing on $Ш(k, E) \times Ш(k, E^\vee)$ is perfect, where $E^\vee \simeq E$. Since $[C] \neq 0$, some $\eta \in Ш(k, E)$ satisfies $\langle [C], \eta \rangle_{\mathrm{CT}} \neq 0$. *This is the only place where finiteness of $Ш$ is used.*

Step 3: the obstruction. Let $\beta \in \mathrm{Br}\, C$ represent $\beta_\eta \in Б(C)$. By the discussion of Manin's group, every $(P_v)_v \in C(\mathbb{A}_k)$ satisfies $\langle (P_v)_v, \beta \rangle = c(\beta_\eta)$, and by Manin's identity $c(\beta_\eta) = \pm \langle [C], \eta \rangle_{\mathrm{CT}} \neq 0$. So no adelic point is orthogonal to β, and $C(\mathbb{A}_k)^{\mathrm{Br}} \subseteq C(\mathbb{A}_k)^{\{\beta\}} = \emptyset$.

Why Manin's identity holds. This is the real content of the theorem, and the proof is a cocycle comparison. The intuition is as follows.

- The class β is *locally* constant everywhere, equal to c_v over k_v. If it were *globally* constant, $\beta = b$ for some $b \in \mathrm{Br}\, k$, then $\sum_v \mathrm{inv}_v\, c_v = \sum_v \mathrm{inv}_v\, b_v = 0$ by reciprocity. So the sum measures the failure of a locally constant class to be globally constant. It is a local-global obstruction, computed by summing local invariants.
- The Cassels–Tate pairing is defined in exactly this way. One lifts the data locally and globally, the mismatch is a family of local Brauer classes that need not come from a global one, and the pairing is the sum of their invariants. Manin checked that, for $\beta = \beta_\eta$ on the torsor C, the two mismatches are the same. □

Back to curves. Let C be a nice curve with a k-rational divisor class of degree 1. Restricting the isomorphism of Proposition 4.3 to locally trivial elements (the same argument works over every k_v) yields

$$Б(C) \simeq Ш(k, J).$$

However, in this case the classes in $Б(C)$ obstruct nothing. If $C(\mathbb{A}_k) \neq \emptyset$, the degree-1 class is represented by a k-rational divisor (see (2.5)), and evaluating a locally constant class on it and using reciprocity shows that its constant value is 0. So for curves of genus $g \geq 2$ with a rational divisor class of degree 1, any Brauer–Manin obstruction must come from classes that are *not* locally constant. This is where the arithmetic of $J(k)$, and not only that of $Ш(k, J)$, enters. We make this precise in Theorem 5.1.

4.4 What is known

Conjecture 3.4 is open in general, but besides genus 0 and the function field analogue discussed after Conjecture 3.4, it is known in the following cases.

- *Genus* 1*, conditionally on* Ш. Let E be the Jacobian of C and suppose $Ш(k, E)$ is finite. If $C(k) = \emptyset$, then C is a non-trivial torsor under E and $C(\mathbb{A}_k)^{\mathrm{Br}} = \emptyset$ by Manin's theorem (Theorem 4.12). If $C(k) \neq \emptyset$, then $C \simeq E$ and $E(\mathbb{A}_k)^{\mathrm{Br}}_{\bullet} = \overline{E(k)}$ by a theorem of Wang [Wan96], which holds for any abelian variety with finite Ш.

- *Jacobians of rank* 0. Scharaschkin [Sch99] showed that if $J(k)$ is finite and $Ш(k, J)$ is finite, then $C(\mathbb{A}_k)^{\mathrm{Br}}_{\bullet} = C(k)$ (see Section 5 for why). Stoll [Sto07] weakened the hypothesis to $Ш(k, J)_{\mathrm{div}} = 0$. He also showed that it is enough for C to admit a non-constant morphism to an abelian variety A with $A(k)$ finite and $Ш(k, A)_{\mathrm{div}} = 0$. Thanks to the results of Gross–Zagier, Kolyvagin and Kato on rank-0 quotients of modular Jacobians, this covers, for example, many modular curves over $\mathbb{Q}$ (see also [HV11]). Related results for Shimura curves can be found in [Sko05, Ara18].

- *Individual curves and numerical evidence.* When $J(k)$ has positive rank, no general result is known. For a given curve, however, the Mordell–Weil sieve [BS10] can often verify the conjecture, or at least the Hasse-principle part of it ($C(\mathbb{A}_k)^{\mathrm{Br}} = \emptyset$ when $C(k) = \emptyset$). Bruin and Stoll [BS08] applied this to all genus-2 curves $y^2 = f(x)$ over $\mathbb{Q}$ with coefficients of f in $\{-3, \ldots, 3\}$. For every such curve without rational points they exhibited an obstruction, conditionally on standard conjectures in a small number of cases. For earlier explicit examples of Brauer–Manin obstructions on curves, see [Fly04]. Poonen's heuristics [Poo06] explain why the sieve should be expected to succeed in general.

5 Computing the Brauer–Manin obstruction on curves

Let C be a nice curve of genus $g \geq 2$ over a number field k with $C(\mathbb{A}_k) \neq \emptyset$. The definition of $C(\mathbb{A}_k)^{\mathrm{Br}}$ involves the whole group $\mathrm{Br}\, C$, and $\mathrm{Br}\, C/\mathrm{Br}_0\, C \simeq \mathrm{H}^1(k, J)/\langle [\mathbf{Pic}^1_{C/k}] \rangle$ is infinite. One cannot hope to write down all of its elements, let alone evaluate them at every adelic point. Nevertheless, the Brauer–Manin obstruction on curves is computable in practice. The point is that all of it can be read off from the Jacobian, and that it can be approximated by finite computations.

5.1 Reduction to the Jacobian

Assume that C has a k-rational divisor class of degree 1, and let $\iota\colon C \to J$ be the corresponding Abel–Jacobi embedding. By Proposition 4.3, $\mathrm{Br}\, C/\mathrm{Br}_0\, C \simeq \mathrm{H}^1(k, J)$. One checks that every such class is the pull-back under ι of an algebraic class on J, coming from $\mathrm{H}^1(k, J^\vee) \subseteq \mathrm{H}^1(k, \mathrm{Pic}\, \overline{J})$ via $J^\vee \simeq J$. Hence

$$(P_v)_v \in C(\mathbb{A}_k)^{\mathrm{Br}} \iff \big(\iota(P_v)\big)_v \text{ is orthogonal to the classes in } \mathrm{Br}\, J \text{ coming from } \mathrm{H}^1(k, J^\vee).$$

Evaluated at local points, such classes give, up to sign, the local Tate pairings

$$J(k_v) \times \mathrm{H}^1(k_v, J^\vee) \longrightarrow \mathbb{Q}/\mathbb{Z}.$$

By Poitou–Tate duality, if $Ш(k, J)$ is finite, then the adelic points of J orthogonal to all of them form exactly $\overline{J(k)}$; this is Wang's theorem mentioned in Section 4.4. Combining the two, one obtains the following description.

Theorem 5.1 (Scharaschkin [Sch99]; Stoll [Sto07])**.** *Let C be a nice curve over a number field k with a k-rational divisor class of degree 1, and suppose that $Ш(k, J)_{\mathrm{div}} = 0$ (e.g. $Ш(k, J)$ finite). Then*

$$C(\mathbb{A}_k)^{\mathrm{Br}}_\bullet = \iota^{-1}\big(\overline{J(k)}\big) = C(\mathbb{A}_k)_\bullet \cap \overline{J(k)},$$

where $\overline{J(k)}$ is the closure of $J(k)$ in $J(\mathbb{A}_k)_\bullet$.

Without any assumption on Ш, Stoll shows that $\overline{J(k)}$ has to be replaced by the image of the profinite Selmer group $\widehat{\mathrm{Sel}}(k, J) \to J(\mathbb{A}_k)_\bullet$. When $Ш(k, J)_{\mathrm{div}} = 0$, the two agree, by the exact sequence $0 \to \widehat{J(k)} \to \widehat{\mathrm{Sel}}(k, J) \to T Ш(k, J) \to 0$. In this form, Conjecture 3.4 for $g \geq 2$ becomes an "adelic Mordell–Lang" statement: *inside $J(\mathbb{A}_k)_\bullet$, the curve $C(\mathbb{A}_k)_\bullet$ meets the closure of the Mordell–Weil group only in $C(k)$*. Note also that Scharaschkin's theorem (the rank-0 case in Section 4.4) is now immediate: if $J(k)$ is finite, then $\overline{J(k)} = J(k)$, and $C(\mathbb{A}_k)_\bullet \cap J(k) = C(k)$.

Finite levels. Theorem 5.1 can be approximated by finite computations. For $n \geq 1$, every element of $\mathrm{Sel}^{(n)}(k, J)$ lies locally in the image of δ_v, so there is a natural map

$$\mathrm{Sel}^{(n)}(k, J) \longrightarrow \prod_v J(k_v)_\bullet / n J(k_v)_\bullet,$$

where $J(k_v)_\bullet = J(k_v)$ if v is non-archimedean and $J(k_v)_\bullet = \pi_0\big(J(k_v)\big)$, the (finite) group of connected components, if v is archimedean. Define

$$C(\mathbb{A}_k)_\bullet^{(n)} = \Big\{(P_v)_v \in C(\mathbb{A}_k)_\bullet \;:\; \big(\iota(P_v) \bmod n\big)_v \in \operatorname{im}\Big(\operatorname{Sel}^{(n)}(k,J) \longrightarrow \prod_v J(k_v)_\bullet/nJ(k_v)_\bullet\Big)\Big\}.$$

Equivalently, $C(\mathbb{A}_k)_\bullet^{(n)}$ is the set of adelic points that lift to an adelic point of some n-covering of C, i.e. of a twist of the pull-back of $[n]\colon J \to J$ along ι. These sets decrease with respect to divisibility of n, and all contain $C(\mathbb{A}_k)_\bullet^{\mathrm{Br}}$, because the obstruction coming from torsors under the finite abelian group scheme $J[n]$ is a Brauer–Manin obstruction [Sko01]. Stoll [Sto07] shows that

$$C(\mathbb{A}_k)_\bullet^{\mathrm{Br}} = \bigcap_{n\geq 1} C(\mathbb{A}_k)_\bullet^{(n)}.$$

In particular, if $C(\mathbb{A}_k)_\bullet^{(n)} = \emptyset$ for some n, then $C(\mathbb{A}_k)^{\mathrm{Br}} = \emptyset$, *unconditionally*. Conversely, if $C(\mathbb{A}_k)^{\mathrm{Br}} = \emptyset$, then $C(\mathbb{A}_k)_\bullet^{(n)} = \emptyset$ for some n: the sets $C(\mathbb{A}_k)_\bullet^{(n)}$ are closed in the compact space $C(\mathbb{A}_k)_\bullet$, and any finitely many of them contain $C(\mathbb{A}_k)_\bullet^{(N)}$ for N the product of their indices. So a Brauer–Manin obstruction to the Hasse principle on a curve is always detected at a finite level.

Decidability. The n-coverings of C with adelic points form a finite, computable set, and each of them can be searched for rational points. Running both procedures in parallel for $n = 1!, 2!, 3!, \ldots$, one either finds a rational point or proves that $C(\mathbb{A}_k)^{\mathrm{Br}} = \emptyset$. So if Conjecture 3.4 holds, there is an algorithm that decides whether $C(k) = \emptyset$ [Sto07]. This algorithm is of theoretical interest only, since computing $\operatorname{Sel}^{(n)}(k,J)$ quickly becomes infeasible as n grows. In practice, small levels already go a long way: two-cover descent [BS09], a variant of the case $n = 2$, shows for many hyperelliptic curves that no two-covering has points everywhere locally, and hence that $C(k) = \emptyset$. For explicit Brauer classes realising such obstructions, see [CS23] for hyperelliptic curves and [BC26] for plane quartics.

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